

The pair singular series of a polynomial IV: the second spectrum

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Abstract

Part III reduced the Cesàro form of the off-diagonal of the pair singular series of an irreducible quadratic f to “pieces” $\mathcal{P}_u(Y)$, one for each way the two roots of f can agree on a part u of the modulus, and showed that everything except the pieces’ “windows” is understood. This paper is about what the pieces actually are. Numerically each piece is $O(\sqrt{Y})$, and $\mathcal{P}_u(Y)/\sqrt{Y}$ oscillates in $\log Y$ at the Laplace eigenvalues of the *even* Maass cusp forms of $\mathrm{SL}_2(\mathbb{Z})$, with amplitudes governed by the values of the forms at the Heegner points, or their integrals over the closed geodesics, of discriminant D : a second spectrum, next to the zeros of ζ_K that govern the diagonal (part II). We explain this by an explicit formula: the piece is a twisted Walfisz sum whose Dirichlet series is a sum over frequencies of Dirichlet series of Salié sums, which the Kuznetsov formula of half-integral weight continues to the half-line with poles at $\frac{1}{2} \pm it_j$, and the Katok–Sarnak correspondence identifies the residues as periods of Maass forms. We prove the formula for a single piece in Riesz-mean form [\[\[target\]\]](#), discuss the uniformity in u that the Cesàro conjecture of part I needs, and report the numerical evidence.

1 Introduction

In plain terms. Parts I and II showed that the main part of the fluctuation of prime values of a polynomial is governed by the zeros of a Dedekind zeta function. The remaining part, the “off-diagonal” of part III, turns out to be governed by a second, entirely different spectrum: the Laplace eigenvalues of the modular surface, the same numbers that govern the distribution of the roots of $x^2 \equiv D \pmod{d}$ (Hooley, Bykovskii, Duke–Friedlander–Iwaniec). Only half of them appear, those of the even forms, and the size of each contribution is the value of the corresponding eigenfunction at a special point, or its integral along a special closed curve, determined by the discriminant of the polynomial. We found this in the data before we had the theory; the theory, once written down, predicts exactly the pattern of presence and absence that the data show.

Set-up. Notation as in part III: $f = t^2 + bt + c$, $D = b^2 - 4c$, $\lambda(p) = p/(p - 2\omega(p))$; for squarefree u with $\omega(u) \geq 1$ and $Q_u(x) = u^2x^2 - D$,

$$\mathcal{P}_u(Y) = \sum_{h \leq Y} (Y-h) (F_u(Q_u(h)) - \mathbb{E}F_u) - \frac{1}{2}(\mathbb{E}F_u - 1)Y, \quad F_u(n) = \sum_{\substack{d|n, d>1 \\ d \text{ admissible}}} \frac{\lambda(d)}{d}, \quad \mathbb{E}F_u = \sum_{\substack{d>1 \\ d \text{ adm}}} \frac{\lambda(d)\omega(d)}{d^2},$$

admissible meaning squarefree, all primes split, coprime to $2Du$. Part III, Theorem A': $\mathrm{Off}_f^*(H) = c_{\mathrm{off}}(f)H + \sum_{u \leq H^{2/3+\varepsilon}} w(u)\mathcal{W}_u(H/u; \log H) + O(H(\log H)^{1-c})$, and the Cesàro form of Conjecture 1 of part I is equivalent to the windows being $o(H \log H)$.

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Results.

- *The spectral formula for a piece* (Theorem 5, [to be proved]): for fixed f and u , and Riesz means of order $m \geq m_0$,

$$\mathcal{P}_u^{(m)}(Y) = c_u Y + Y^{1/2} \sum_j (\alpha_j(u, f) Y^{it_j} + \overline{\alpha_j(u, f)} Y^{-it_j}) \gamma_m(t_j) + O(Y^{1/2-\delta}),$$

the sum over the even Maass cusp forms u_j of $\mathrm{SL}_2(\mathbb{Z})$ with Laplace eigenvalue $\frac{1}{4} + t_j^2$ (and, for $u > 1$, the newforms of the levels dividing u^2), with $\alpha_j(u, f)$ an explicit multiple of the Katok–Sarnak period of u_j at discriminant D , and $\gamma_m(t) \ll |t|^{-m-1}$.

- *Numerical confirmation* (§6): for five quadratics the even parameters explain the variance of $\mathcal{P}_u(Y)/\sqrt{Y}$ at the 92nd–100th percentile among random frequency sets, the odd ones at the 12th–66th; the dominant line $t_1 = 13.7798$ has stable amplitude and phase over two decades of $\log Y$; the two polynomials with exponentially small periods ($D = -8, -163$) show nothing.
- *What the Cesàro conjecture needs* (§7): the sum of the pieces over $u \leq H^{2/3}$ converges to $c_{\mathrm{off}} H + o(H)$ provided the spectral formula holds with an error $O((H/u)^{1/2+\varepsilon} u^A)$, $A < \frac{1}{4}$; the level dependence of the Kuznetsov formula is therefore the whole remaining problem.

2 The piece as a twisted Walfisz sum

Throughout, f is an irreducible quadratic of discriminant D and u is squarefree with $\omega(u) \geq 1$. Call a modulus d *admissible* (for u) if it is squarefree, coprime to $2Du$, and every prime dividing it is split, i.e. $\omega(p) = 2$. For admissible d put

$$R_d = R_d^{(u)} = \{x \bmod d : u^2 x^2 \equiv D\}, \quad \rho(d) = |R_d| = \omega(d) = 2^{\nu(d)}, \quad \rho_k(d) = \sum_{x \in R_d} e\left(\frac{kx}{d}\right),$$

ν the number of prime factors and $e(y) = e^{2\pi i y}$. Two properties are used constantly: R_d is *symmetric*, $x \in R_d \Rightarrow -x \in R_d$, and $0 \notin R_d$, both because $(D, d) = 1$; consequently every $\rho_k(d)$ is real. Write $\lambda(p) = p/(p-4)$ on split primes, λ multiplicative, and

$$F_u(n) = \sum_{\substack{d|n, d>1 \\ d \text{ admissible}}} \frac{\lambda(d)}{d}, \quad \mathbb{E}F_u = \sum_{\substack{d>1 \\ d \text{ admissible}}} \frac{\lambda(d)\rho(d)}{d^2}, \quad Q_u(h) = u^2 h^2 - D, \quad (1)$$

both sums absolutely convergent ($\lambda(d)\rho(d) \ll d^\varepsilon$). *All sums over moduli below run over admissible $d > 1$* : including $d = 1$ would add 1 to F_u and to $\mathbb{E}F_u$, leaving (2) unchanged but spoiling the symmetry used in Lemma 1, since $R_1 = \{0\}$ is not a union of pairs $\{x, -x\}$ with $x \neq -x$. The *sharp piece* and the *Riesz pieces* are

$$S_u(t) = \sum_{h \leq t} (F_u(Q_u(h)) - \mathbb{E}F_u), \quad \mathcal{P}_u^{(m)}(Y) = \frac{1}{m!} \sum_{h \leq Y} (Y - h)^m (F_u(Q_u(h)) - \mathbb{E}F_u) \quad (m \geq 0), \quad (2)$$

so that $\mathcal{P}_u^{(0)} = S_u$ and, for integers Y , $\mathcal{P}_u^{(1)}(Y) = \sum_{t < Y} S_u(t)$; up to the explicit constant of part III, Remark ??, $\mathcal{P}_u^{(1)}(Y)$ is the piece $\mathcal{P}_u(Y)$ of part III. The connection with part III is that $F_u(Q_u(h))$ is the divisor-sum form of the inner sum of a piece: $d \mid Q_u(h)$ exactly when h lies in one of the $\rho(d)$ classes R_d .

Lemma 1 (sawtooth form). *Let $\psi(y) = \frac{1}{2} - \{y\}$. For every $t \geq 0$,*

$$S_u(t) = \sum_{d \text{ admissible}} \frac{\lambda(d)}{d} \sum_{x \in R_d} \psi\left(\frac{t-x}{d}\right), \quad (3)$$

the series converging absolutely.

Proof. The right-hand side is not absolutely convergent term by term, since $\sum_d \lambda(d)\rho(d)/d$ diverges; it is summed with the terms x and $-x$ grouped, and converges in that sense, as shown at the end. Expanding F_u and counting,

$$\sum_{h \leq t} F_u(Q_u(h)) = \sum_d \frac{\lambda(d)}{d} \#\{h \leq t : d \mid Q_u(h)\} = \sum_d \frac{\lambda(d)}{d} \sum_{x \in R_d} \#\{h \leq t : h \equiv x \pmod{d}\},$$

the interchange being legitimate because $\sum_{d \mid Q_u(h)} \lambda(d)/d$ converges absolutely for each $h \leq t$ and there are finitely many h . Represent each $x \in R_d$ by $\langle x \rangle_d \in [1, d]$. For $0 \leq t$ and $1 \leq x \leq d$,

$$\#\{1 \leq h \leq t : h \equiv x \pmod{d}\} = \left\lfloor \frac{t-x}{d} \right\rfloor + 1,$$

valid also when $x > t$, where both sides vanish because $-1 < (t-x)/d < 0$. Writing $\lfloor y \rfloor = y - \frac{1}{2} + \psi(y)$,

$$\#\{h \leq t : h \equiv x \pmod{d}\} = \frac{t}{d} + \left(\frac{1}{2} - \frac{x}{d}\right) + \psi\left(\frac{t-x}{d}\right).$$

Now sum over $x \in R_d$. Since R_d is symmetric and $0 \notin R_d$, its representatives split into pairs $\{\langle x \rangle_d, d - \langle x \rangle_d\}$, and $(\frac{1}{2} - \frac{x}{d}) + (\frac{1}{2} - \frac{d-x}{d}) = 0$; hence $\sum_{x \in R_d} (\frac{1}{2} - \langle x \rangle_d/d) = 0$ for every admissible $d > 1$. Therefore

$$\sum_{h \leq t} F_u(Q_u(h)) = t \sum_d \frac{\lambda(d)\rho(d)}{d^2} + \sum_d \frac{\lambda(d)}{d} \sum_{x \in R_d} \psi\left(\frac{t-x}{d}\right) = t \mathbb{E}F_u + \sum_d \frac{\lambda(d)}{d} \sum_{x \in R_d} \psi\left(\frac{t-x}{d}\right),$$

which is (3). For the convergence, group the terms of a pair $\{x, -x\}$ and write $a = \langle x \rangle_d$ with $a \leq d/2$, so that the pair is $\{a, d-a\}$. If $d > 2t$ and $a > t$ then $\{(t-a)/d\} = (t-a+d)/d$ and $\{(t+a)/d\} = (t+a)/d$, so by the periodicity of ψ the two sawtooths sum to $-2t/d$; if $a \leq t$ their sum is bounded by 1, and this happens for at most $\rho(d)$ values with $a \leq t < d/2$, a set that is empty once $d > 2t$. Hence for $d > 2t$ each pair contributes $\ll t/d$ and

$$\sum_{d > 2t} \frac{\lambda(d)}{d} \cdot \frac{t\rho(d)}{d} \ll t \sum_{d > 2t} \frac{\lambda(d)\rho(d)}{d^2} \ll t^\varepsilon,$$

while the range $d \leq 2t$ is a finite sum. So the paired series converges and (3) holds. $\square$

In words. A piece counts, for each admissible modulus d , how far the number of $h \leq t$ with $d \mid Q_u(h)$ departs from its expectation, weighted by $\lambda(d)/d$; and that departure is a sawtooth evaluated at the roots. The linear term that normally accompanies such a count cancels because the roots come in pairs $\pm x$. This is the same cancellation that makes the diagonal of part I a sum of sawtooths $\{H/d\} - \frac{1}{2}$, and it is why the piece is a Walfisz-type sum: in the classical case $\sum_{n \leq x} n^{-1}(\frac{1}{2} - \{x/n\})$ one sums a sawtooth over all moduli with weight $1/n$, and here one sums it over admissible moduli, at the roots of a quadratic congruence, with weight $\lambda(d)/d$.

Lemma 2 (Weyl-sum form). *For every $t \geq 0$ and every $K \geq 1$,*

$$S_u(t) = \frac{1}{\pi} \sum_{k \geq 1} \frac{1}{k} \sum_{d \text{ admissible}} \frac{\lambda(d)\rho_k(d)}{d} \sin \frac{2\pi kt}{d}, \quad (4)$$

the k -sum converging boundedly. Consequently, for the Riesz pieces of order $m \geq 1$ and integer Y ,

$$\mathcal{P}_u^{(m)}(Y) = \frac{1}{\pi} \sum_{k \geq 1} \frac{1}{k} \sum_d \frac{\lambda(d) \rho_k(d)}{d} \Phi_m\left(\frac{2\pi k}{d}; Y\right), \quad \Phi_m(\theta; Y) = \frac{1}{m!} \sum_{h \leq Y} (Y - h)^m \sin(\theta h), \quad (5)$$

with $\Phi_m(\theta; Y) \ll_m \min(Y^{m+1}, Y^m \|\theta/2\pi\|^{-1})$, $\|\cdot\|$ the distance to $\mathbb{Z}$.

Proof. The Fourier expansion $\psi(y) = \sum_{k \geq 1} \sin(2\pi k y)/(\pi k)$ holds for $y \notin \mathbb{Z}$, boundedly and uniformly on compact subsets of $\mathbb{R} \setminus \mathbb{Z}$, and the partial sums are uniformly bounded. Insert it into (3) and use, for each d ,

$$\sum_{x \in R_d} \sin \frac{2\pi k(t-x)}{d} = \Im \left[e\left(\frac{kt}{d}\right) \overline{\rho_k(d)} \right] = \rho_k(d) \sin \frac{2\pi kt}{d},$$

since $\rho_k(d)$ is real. The interchange of the k -sum with the d -sum is justified by the uniform boundedness of the partial sums of the Fourier series together with the absolute convergence established in Lemma 1 (dominated convergence for series). For (5), apply $\frac{1}{m!} \sum_{h \leq Y} (Y - h)^m$ to $t = h$ in (4), which is a finite sum, and evaluate Φ_m by summation by parts m times against the geometric sum $\sum_{h \leq Y} e(\theta h/2\pi) \ll \min(Y, \|\theta/2\pi\|^{-1})$. $\square$

Remark 3. Two features of (5) decide everything that follows. First, the frequencies k enter only through the Weyl sums $\rho_k(d)$ of the dilated roots, so the arithmetic of the problem is entirely contained in the Dirichlet series

$$W_k(s) = \sum_{d \text{ admissible}} \frac{\lambda(d) \rho_k(d)}{d^s} \quad (\Re s > 1), \quad (6)$$

one for each k . Second, the weight $1/k$ in front decays only like k^{-1} , so a bound for W_k that loses a power of k is expensive: this is the same phenomenon that forced the condition $\theta + 6B < 1$ in part III, Theorem A, and it is the reason we do not bound the $\rho_k(d)$ one frequency at a time. The route of this paper is instead to continue $W_k(s)$ analytically, where the dependence on k appears as a Fourier coefficient of an automorphic form and therefore *decays* on average.

Proposition 4 (the Mellin representation). *Let $m \geq 1$ and let*

$$A_u(s) = \sum_{h \geq 1} \frac{F_u(Q_u(h)) - \mathbb{E}F_u}{h^s} \quad (\Re s > 1).$$

Then A_u extends to a function holomorphic in $\Re s > \frac{1}{2}$ except possibly for the singularities inherited from the W_k , and for integer $Y \geq 2$ and any $c > 1$,

$$\mathcal{P}_u^{(m)}(Y) = \frac{1}{2\pi i} \int_{(c)} A_u(s) \frac{Y^{s+m}}{s(s+1) \cdots (s+m)} ds. \quad (7)$$

Moreover, for $0 < \Re s < 1$,

$$A_u(s) = \frac{2\Gamma(1-s)}{(2\pi)^{1-s}} \sin \frac{\pi s}{2} \sum_{k \geq 1} k^{s-1} W_k(s+1) + (\text{entire}), \quad (8)$$

whenever the k -sum converges, where W_k is as in (6).

Proof. (7) is the Riesz–Perron formula, valid because $F_u(Q_u(h)) - \mathbb{E}F_u \ll h^\varepsilon$ so that A_u converges absolutely for $\Re s > 1$. For (8), expand F_u and sum over each residue class:

$$\sum_{h \geq 1} \frac{F_u(Q_u(h))}{h^s} = \sum_d \frac{\lambda(d)}{d^{1+s}} \sum_{x \in R_d} \zeta\left(s, \frac{\langle x \rangle_d}{d}\right),$$

$\zeta(s, \alpha)$ the Hurwitz zeta function. Since $\zeta(s, \alpha)$ has residue 1 at $s = 1$ for every α , the pole of the d -th term is $\rho(d)\lambda(d)d^{-1-s}$, and summing over d these residues total $\mathbb{E}F_u$, which is exactly the pole of $\mathbb{E}F_u\zeta(s)$: each bracket $\lambda(d)d^{-1-s} \sum_x \zeta(s, \langle x \rangle_d/d) - \lambda(d)\rho(d)d^{-2}\zeta(s)$ is entire, which is the statement that A_u continues past $s = 1$. Hurwitz’s formula,

$$\zeta(s, \alpha) = \frac{2\Gamma(1-s)}{(2\pi)^{1-s}} \left[\sin \frac{\pi s}{2} \sum_{k \geq 1} \frac{\cos 2\pi k \alpha}{k^{1-s}} + \cos \frac{\pi s}{2} \sum_{k \geq 1} \frac{\sin 2\pi k \alpha}{k^{1-s}} \right],$$

valid for $\Re s < 0$ and by continuation for $\Re s < 1$ with the series interpreted as the analytic continuations of the periodic zeta functions, gives after summing over the symmetric set R_d

$$\sum_{x \in R_d} \zeta\left(s, \frac{\langle x \rangle_d}{d}\right) = \frac{2\Gamma(1-s)}{(2\pi)^{1-s}} \sin \frac{\pi s}{2} \sum_{k \geq 1} \frac{\rho_k(d)}{k^{1-s}},$$

the sine term vanishing identically because $\sum_{x \in R_d} \sin(2\pi k x/d) = 0$ by symmetry. Multiplying by $\lambda(d)d^{-1-s}$ and summing over d gives (8). $\square$

In words. The piece is determined by one Dirichlet series for each frequency, namely the Dirichlet series of the Weyl sums of the roots of $u^2x^2 \equiv D$; the symmetry of the root set kills the odd half of Hurwitz’s formula, so only the even half survives. That the odd half disappears here is the first appearance of the parity phenomenon that will select the even Maass forms in §4, and it has the same source: the root set of a quadratic congruence is invariant under $x \mapsto -x$.

3 Salié sums and the Kuznetsov formula of weight one half

[Weyl sums of quadratic roots as Salié sums (DFI 1995 §2); the dilation as a fractional frequency; Kuznetsov for the theta multiplier on $\Gamma_0(4)$ (Proskurin) and its use by Bykovskii 1984 and DFI 2012; continuation of $W_k(s)$ to $\Re s > \frac{1}{2}$; the admissibility sieve; the twist $\lambda = 1 * \kappa$.]

4 Katok–Sarnak and the parity rule

[Statement of the Katok–Sarnak correspondence; residues as periods; vanishing for odd forms at i , ρ , over conjugate pairs of Heegner points and reflection-symmetric geodesics; the level $4u^2$ and oldforms.]

5 The spectral formula for a piece

Theorem 5 ([target]). [statement as in the introduction, with the precise hypotheses on u and m_0 .]

6 Numerical evidence

[Table of NOTES-routeR.md §1; figure of $\mathcal{P}_1(Y)/\sqrt{Y}$ for $t^2 - 2$ against the fitted even lines; stability of amplitudes and phases; scripts.]

7 Uniformity in u and the Cesàro conjecture

[The condition $A < \frac{1}{4}$; what the spectral large sieve gives; the windows of part III in spectral terms.]

AI-assisted research: disclosure

This work was carried out with heavy use of large language models, the Anthropic models Claude Fable 5.1 and Claude Opus 5, used interactively by the author in September 2026 as research assistants. The author posed the questions, set the direction and the scope, and made every decision about what to pursue, claim and submit. The model was used to draft mathematical arguments, to write and run the software behind every number in the paper, to search the literature, and to draft the text; all of this was done in dialogue with the author, who verified the code and the numerical results, reviewed the proofs, and takes full responsibility for the content. The model is not an author. The complete record of the work is in the public repository <https://github.com/gewure/ulamnd> (directory `research/`).

References